

T3 UV one-loop RGE term comparison

Each running coupling has its own table. Rows are model configurations and columns are additive structures in the beta function. Every populated cell contains the complete model-specific term, including its numerical coefficient. A dash means that the term is absent for that configuration. All equations use $16\pi^2 \mu \, dX/d\mu = \beta_X^{(1)}$. Gauge-coupling results use the same RGBeta $g^2 \rightarrow g$ conversion as in the previous report.

$$\beta_{g_Y}^{(1)}$$

$$16\pi^2 \mu \frac{dg_Y}{d\mu} = \beta_{g_Y}^{(1)}$$

Column Representative term structure	
T1	$\frac{59g_Y^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{59g_Y^3}{6}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{17g_Y^3}{2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\frac{21g_Y^3}{2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{55g_Y^3}{6}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{59g_Y^3}{6}$

$\beta_{g_2}^{(1)}$

$16\pi^2\mu\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$

Column Representative term structure	
T1	$-\frac{g_2^3}{2}$
T2	$-\frac{17g_2^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{g_2^3}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-\frac{17g_2^3}{6}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\frac{31g_2^3}{6}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{g_2^3}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{g_2^3}{6}$	—

$$\beta_{g_3}^{(1)}$$

$$16\pi^2\,\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Column Representative term structure

T1	$-7g_3^3$
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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-7g_3^3$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-7g_3^3$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-7g_3^3$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$

$\beta_{Y_u}^{(1)}$

$16\pi^2\mu\frac{dY_u}{d\mu}=\beta_{Y_u}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$
T2	$+\text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$
T3	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+\text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+\text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+\text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+\text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+\text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$
T3	$-8g_3^2\ (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{17}{12}gY^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17}{12}gY^2(Y_u)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{17}{12}gY^2(Y_u)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{17}{12}gY^2(Y_u)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{17}{12}gY^2(Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{17}{12}gY^2(Y_u)_{ij}$

$\beta_{Y_d}^{(1)}$

$16\pi^2\,\mu\,\frac{dY_d}{d\mu}=\beta_{Y_d}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$
T2	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$
T3	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$
T3	$-8g_3^2\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12}gY^2(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{5}{12}gY^2(Y_d)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{5}{12}gY^2(Y_d)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{5}{12}gY^2(Y_d)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{5}{12}gY^2(Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{5}{12}gY^2(Y_d)_{ij}$

$\beta_{Y_e}^{(1)}$

$16\pi^2\,\mu\,\frac{dY_e}{d\mu}=\beta_{Y_e}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$
T2	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$
T3	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$
T2	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$
T3	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+2\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$

Terms 3 of 3

Column	Representative term structure
T1	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$

$\beta_{y_1}^{(1)}$

$16\pi^2\mu\frac{dy_1}{d\mu}=\beta_{y_1}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$
T2	$+\frac{3}{4}\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}$
T3	$+\frac{1}{2}\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$	$+\frac{3}{4}\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}$	$+\frac{1}{2}\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

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Column	Representative term structure
T1	$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$
T2	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$
T3	$-\frac{3}{2} g_Y^2 \left(y_1 \right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-\frac{9}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{33}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_Y^2 \left(y_1 \right)_{ij}$

Terms 3 of 4

Column	Representative term structure
T1	$\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{i j}$
T2	$+\frac{1}{2}\left(\left(y_2\left(y_2\right)^{\dagger} y_1\right)_{i j}+\left(Y_e\left(Y_e\right)^{\dagger} y_1\right)_{i j}+3\left(y_1\left(y_1\right)^{\dagger} y_1\right)_{i j}\right)$
T3	$\frac{3}{2} \text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{i j}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{i j}$	$+\frac{1}{2}\left(\left(y_2\left(y_2\right)^{\dagger} y_1\right)_{i j}+\left(Y_e\left(Y_e\right)^{\dagger} y_1\right)_{i j}+3\left(y_1\left(y_1\right)^{\dagger} y_1\right)_{i j}\right)$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$\frac{3}{2} \text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{i j}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{i j}$	$+\frac{1}{2}\left(\left(y_2\left(y_2\right)^{\dagger} y_1\right)_{i j}+\left(Y_e\left(Y_e\right)^{\dagger} y_1\right)_{i j}+3\left(y_1\left(y_1\right)^{\dagger} y_1\right)_{i j}\right)$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{i j}$	—	—

Terms 4 of 4

Column	Representative term structure
T1	$+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+6\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$

$\beta_{y_2}^{(1)}$

$16\pi^2\mu\frac{dy_2}{d\mu}=\beta_{y_2}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$
T2	$+\frac{1}{2}\Big((y_1\ (y_1)^\dagger\ y_2)_{ij}+\Big(Y_e\ (Y_e)^\dagger\ y_2\Big)_{ij}+3\Big(y_2\ (y_2)^\dagger\ y_2\Big)_{ij}\Big)$
T3	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	$+\frac{1}{2}\Big((y_1\ (y_1)^\dagger\ y_2)_{ij}+\Big(Y_e\ (Y_e)^\dagger\ y_2\Big)_{ij}+3\Big(y_2\ (y_2)^\dagger\ y_2\Big)_{ij}\Big)$	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	$+\frac{1}{2}\Big((y_1\ (y_1)^\dagger\ y_2)_{ij}+\Big(Y_e\ (Y_e)^\dagger\ y_2\Big)_{ij}+3\Big(y_2\ (y_2)^\dagger\ y_2\Big)_{ij}\Big)$	$-\frac{9}{4}g_2^2\,(y_2)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{33}{4}g_2^2\,(y_2)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	—	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$

Terms 2 of 4

Column	Representative term structure
T1	$-\frac{3}{2}g_Y^2(y_2)_{ij}$
T2	$\frac{3}{2}\text{Tr}\left(y_2(y_2)^\dagger\right)(y_2)_{ij}$
T3	$+\frac{1}{4}(3\left(y_1(y_1)^\dagger y_2\right)_{ij}+2\left(Y_e(Y_e)^\dagger y_2\right)_{ij}+5\left(y_2(y_2)^\dagger y_2\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}g_Y^2(y_2)_{ij}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3}{2}g_Y^2(y_2)_{ij}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{3}{2}g_Y^2(y_2)_{ij}$	$\frac{3}{2}\text{Tr}\left(y_2(y_2)^\dagger\right)(y_2)_{ij}$	$+\frac{1}{4}(3\left(y_1(y_1)^\dagger y_2\right)_{ij}+2\left(Y_e(Y_e)^\dagger y_2\right)_{ij}+5\left(y_2(y_2)^\dagger y_2\right)_{ij})$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3}{2}g_Y^2(y_2)_{ij}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}g_Y^2(y_2)_{ij}$	—	$+\frac{1}{4}(3\left(y_1(y_1)^\dagger y_2\right)_{ij}+2\left(Y_e(Y_e)^\dagger y_2\right)_{ij}+6\left(y_2(y_2)^\dagger y_2\right)_{ij})$

Terms 3 of 4

Column	Representative term structure
T1	$2 \operatorname{Tr} \left(y_2 \left(y_2 \right)^{\dagger} \right) \left(y_2 \right)_{ij}$
T2	$+ \frac{3}{4} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \right)_{ij}$
T3	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_2 \right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$2 \operatorname{Tr} \left(y_2 \left(y_2 \right)^{\dagger} \right) \left(y_2 \right)_{ij}$	$+ \frac{3}{4} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \right)_{ij}$	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_2 \right)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 4

Column	Representative term structure
T1	$+ \left(y_2 \left(y_2 \right)^{\dagger} y_2 \right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+ \left(y_2 \left(y_2 \right)^{\dagger} y_2 \right)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{M_F}^{(1)}$

$16\pi^2\,\mu\,\frac{dM_F}{d\mu}=\beta_{M_F}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+3\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$
T2	$-\frac{9}{2}g_2^2\,(M_F)_{ij}$
T3	$-\frac{3}{2}g_Y^2\,(M_F)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+3\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	$-\frac{9}{2}g_2^2\,(M_F)_{ij}$	$-\frac{3}{2}g_Y^2\,(M_F)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-\frac{3}{2}g_Y^2\,(M_F)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{3}{2}g_Y^2\,(M_F)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{1}{4}(3\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+2\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	$-\frac{9}{2}g_2^2\,(M_F)_{ij}$	$-\frac{3}{2}g_Y^2\,(M_F)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{9}{2}g_2^2\,(M_F)_{ij}$	$-\frac{3}{2}g_Y^2\,(M_F)_{ij}$

Terms 2 of 3

Column	Representative term structure
T1	$\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$
T2	$+ \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$
T3	$\frac{1}{2}((y_1)^T\ (y_1)^*\ M_F)_{ij} + \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$	$+ \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$\frac{1}{2}((y_1)^T\ (y_1)^*\ M_F)_{ij} + \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\frac{3}{4}((y_1)^T\ (y_1)^*\ M_F)_{ij} + \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$

Terms 3 of 3

Column	Representative term structure
T1	$-12g_2^2(M_F)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-12g_2^2(M_F)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{m^2_{S_1}}^{(1)}$

$16\pi^2\mu\frac{dm^2_{S_1}}{d\mu}=\beta_{m^2_{S_1}}^{(1)}$

Terms 1 of 3

Column		Representative term structure
T1	$m^2_{S_1}$	$(4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+4\lambda_{S_1})$
T2	$-8\text{Tr}\left(M_F\left(M_F\right)^{\dagger}\left(y_1\right)^T\left(y_1\right)^{*}\right)$	
T3	$+6\lambda_{12}m^2_{S_2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$m^2_{S_1}(4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+4\lambda_{S_1})$	$-8\text{Tr}\left(M_F\left(M_F\right)^{\dagger}\left(y_1\right)^T\left(y_1\right)^{*}\right)$	$+6\lambda_{12}m^2_{S_2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$m^2_{S_1}(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+6\lambda_{S_1})$	$-4\text{Tr}\left(M_F\left(M_F\right)^{\dagger}\left(y_1\right)^T\left(y_1\right)^{*}\right)$	$+4\lambda_{12}m^2_{S_2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$m^2_{S_1}(3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+6\lambda_{S_1})$	$-6\text{Tr}\left(M_F\left(M_F\right)^{\dagger}\left(y_1\right)^T\left(y_1\right)^{*}\right)$	$+4\lambda_{12}m^2_{S_2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-4\text{Tr}\left(M_F\left(M_F\right)^{\dagger}\left(y_1\right)^T\left(y_1\right)^{*}\right)$	$+2\lambda_{12}m^2_{S_2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-4\text{Tr}\left(M_F\left(M_F\right)^{\dagger}\left(y_1\right)^T\left(y_1\right)^{*}\right)$	—

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{9g_2^2m_{S_1}^2}{2}$
T2	$m_{S_1}^2(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}})$
T3	$-12g_2^2m_{S_1}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{9g_2^2m_{S_1}^2}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{9g_2^2m_{S_1}^2}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$m_{S_1}^2(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}})$	$-12g_2^2m_{S_1}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$m_{S_1}^2(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}})$	$-12g_2^2m_{S_1}^2$

Terms 3 of 3

Column	Representative term structure
T1	$+m_{S_2}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+m_{S_2}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

$\beta_{m_{S_2}^2}^{(1)}$

$16\pi^2\mu\frac{dm_{S_2}^2}{d\mu}=\beta_{m_{S_2}^2}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$m_{S_2}^2(2\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$
T2	$-4\text{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)$
T3	$-12g_2^2m_{S_2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$m_{S_2}^2(2\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$	$-4\text{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)$	$-12g_2^2m_{S_2}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-4\text{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-6\text{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-8\text{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$m_{S_2}^2(2\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$	$-4\text{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)$	$-12g_2^2m_{S_2}^2$

Terms 2 of 3

Column	Representative term structure
T1	$-6g_Y^2 m_{S_2}^2$
T2	$+2\lambda_{12} m_{S_1}^2$
T3	$m_{S_2}^2 (2 \operatorname{Tr} (y_2 \ (y_2)^\dagger) + 6\lambda_{S_2})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-6g_Y^2 m_{S_2}^2$	$+2\lambda_{12} m_{S_1}^2$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-6g_Y^2 m_{S_2}^2$	$+4\lambda_{12} m_{S_1}^2$	$m_{S_2}^2 (2 \operatorname{Tr} (y_2 \ (y_2)^\dagger) + 6\lambda_{S_2})$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-6g_Y^2 m_{S_2}^2$	$+4\lambda_{12} m_{S_1}^2$	$m_{S_2}^2 (3 \operatorname{Tr} (y_2 \ (y_2)^\dagger) + 6\lambda_{S_2})$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-6g_Y^2 m_{S_2}^2$	$+6\lambda_{12} m_{S_1}^2$	$m_{S_2}^2 (4 \operatorname{Tr} (y_2 \ (y_2)^\dagger) + 4\lambda_{S_2})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-6g_Y^2 m_{S_2}^2$	—	—

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{9g_2^2m_{S_2}^2}{2}$
T2	$+m_{S_1}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{9g_2^2m_{S_2}^2}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{9g_2^2m_{S_2}^2}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+m_{S_1}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

$\beta_{\lambda_H}^{(1)}$

$16\pi^2\mu\frac{d\lambda_H}{d\mu}=\beta_{\lambda_H}^{(1)}$

Terms 1 of 6

Column	Representative term structure	
T1	$\lambda_H(12\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+4\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+12\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	
T2	$-12\text{Tr}\big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\big)$	
T3	$-4\text{Tr}\big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_H(12\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+4\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+12\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-12\text{Tr}\big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\big)$	$-4\text{Tr}\big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\big)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\lambda_H(12\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+4\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+12\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-12\text{Tr}\big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\big)$	$-4\text{Tr}\big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\big)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\lambda_H(12\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+4\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+12\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-12\text{Tr}\big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\big)$	$-4\text{Tr}\big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\big)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\lambda_H(12\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+4\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+12\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-12\text{Tr}\big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\big)$	$-4\text{Tr}\big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\big)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_H(12\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+4\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+12\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-12\text{Tr}\big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\big)$	$-4\text{Tr}\big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\big)$

Terms 2 of 6

Column	Representative term structure
T1	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$
T2	$+4\lambda_{T3}\left(\lambda_{T3}\right)^{*}$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+\frac{9g_2^4}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+\frac{9g_2^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+\frac{9g_2^4}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+\frac{9g_2^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+\frac{9g_2^4}{4}$

Terms 3 of 6

Column	Representative term structure
T1	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$
T2	$-9g_2^2\lambda_H$
T3	$+\frac{3g_Y^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$

Terms 4 of 6

Column	Representative term structure
T1	$+12\lambda_H^2$
T2	$+2\lambda_{H1}^2$
T3	$+6\lambda_{H2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+12\lambda_H^2$	$+2\lambda_{H1}^2$	$+6\lambda_{H2}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+4\lambda_{H2}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+4\lambda_{H2}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	$+2\lambda_{H2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	$+6\lambda_{H2}^2$

Terms 5 of 6

Column	Representative term structure
T1	$+3\lambda_{H2}^{\text{Adj}^2}$
T2	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$
T3	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	—	—

Terms 6 of 6

Column	Representative term structure
T1	$+3\lambda_{H1}^{\text{Adj}^2}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}^2}$

$\beta_{\lambda_{S_1}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_1}}{d\mu}=\beta_{\lambda_{S_1}}^{(1)}$

Terms 1 of 6

Column	Representative term structure
T1	$8\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$
T2	$-8\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$
T3	$+6\lambda_{12}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$8\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-8\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+6\lambda_{12}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$4\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+4\lambda_{12}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$6\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-5\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+4\lambda_{12}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+2\lambda_{12}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+6\lambda_{12}^2$

Terms 2 of 6

Column	Representative term structure
T1	$+4\lambda_{H1}^2$
T2	$+10\lambda_{S1}^2$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H1}^2$	$+10\lambda_{S1}^2$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{H1}^2$	$+12\lambda_{S1}^2$	$+\frac{9g_2^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{H1}^2$	$+12\lambda_{S1}^2$	$+\frac{9g_2^4}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+4\lambda_{H1}^2$	$+14\lambda_{S1}^2$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H1}^2$	$+14\lambda_{S1}^2$	—

Terms 3 of 6

Column	Representative term structure
T1	$-9g_2^2\lambda_{S_1}$
T2	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$
T3	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-9g_2^2\lambda_{S_1}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-9g_2^2\lambda_{S_1}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 6

Column	Representative term structure
T1	$\lambda_{S_1} (4 \operatorname{Tr} (y_1 \ (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$
T2	$+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} (y_1 \ (y_1)^\dagger)$
T3	$+21 g_2^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\lambda_{S_1} (4 \operatorname{Tr} (y_1 \ (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$	$+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} (y_1 \ (y_1)^\dagger)$	$+21 g_2^4$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_1} (4 \operatorname{Tr} (y_1 \ (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$	$+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} (y_1 \ (y_1)^\dagger)$	$+21 g_2^4$

Terms 5 of 6

Column	Representative term structure
T1	$+g_2^2(-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$
T2	$+\frac{3\lambda_{H_1}^{\text{Adj}^2}}{2}$
T3	$+26\lambda_{S_1}^{\text{Adj}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+g_2^2(-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$	$+\frac{3\lambda_{H_1}^{\text{Adj}^2}}{2}$	$+26\lambda_{S_1}^{\text{Adj}^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_2^2(-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$	$+\frac{3\lambda_{H_1}^{\text{Adj}^2}}{2}$	$+26\lambda_{S_1}^{\text{Adj}^2}$

Terms 6 of 6

Column	Representative term structure
T1	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$
T2	$+6\lambda_{12}^{\text{Adj}^2}$
T3	$+6\lambda_{12}^{\text{Cross}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$	$+6\lambda_{12}^{\text{Adj}^2}$	$+6\lambda_{12}^{\text{Cross}^2}$

$\beta_{\lambda_{S_2}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_2}}{d\mu}=\beta_{\lambda_{S_2}}^{(1)}$

Terms 1 of 8

Column	Representative term structure
T1	$\lambda_{S_2}\left(4\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+14\lambda_{S_2}^{\text{Adj}}\right)$
T2	$+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$
T3	$-3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}\left(4\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+14\lambda_{S_2}^{\text{Adj}}\right)$	$+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$	$-3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-4\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-5\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$-8\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}\left(4\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+14\lambda_{S_2}^{\text{Adj}}\right)$	$+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$	$-3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$

Terms 2 of 8

Column	Representative term structure
T1	$+21g_2^4$
T2	$+g_Y^2\left(12g_2^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$
T3	$+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+21g_2^4$	$+g_Y^2\left(12g_2^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$	$+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+21g_2^4$	$+g_Y^2\left(12g_2^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$	$+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$

Terms 3 of 8

Column	Representative term structure
T1	$+12g_Y^4$
T2	$+2\lambda_{12}^2$
T3	$+4\lambda_{H2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+12g_Y^4$	$+2\lambda_{12}^2$	$+4\lambda_{H2}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+12g_Y^4$	$+4\lambda_{12}^2$	$+4\lambda_{H2}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+12g_Y^4$	$+4\lambda_{12}^2$	$+4\lambda_{H2}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+12g_Y^4$	$+6\lambda_{12}^2$	$+4\lambda_{H2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12g_Y^4$	$+6\lambda_{12}^2$	$+4\lambda_{H2}^2$

Terms 4 of 8

Column	Representative term structure
T1	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$
T2	$+14\lambda_{S_2}^2$
T3	$+26\lambda_{S_2}^{\text{Adj}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+14\lambda_{S_2}^2$	$+26\lambda_{S_2}^{\text{Adj}^2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$+12\lambda_{S_2}^2$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+12\lambda_{S_2}^2$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$+10\lambda_{S_2}^2$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+14\lambda_{S_2}^2$	$+26\lambda_{S_2}^{\text{Adj}^2}$

Terms 5 of 8

Column	Representative term structure
T1	$4\lambda_{S_2} \text{Tr}\left(y_2 \left(y_2\right)^\dagger\right)$
T2	$+\frac{9g_2^4}{4}$
T3	$+g_Y^2\left(6g_2^2-12\lambda_{S_2}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$4\lambda_{S_2} \text{Tr}\left(y_2 \left(y_2\right)^\dagger\right)$	$+\frac{9g_2^4}{4}$	$+g_Y^2\left(6g_2^2-12\lambda_{S_2}\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$6\lambda_{S_2} \text{Tr}\left(y_2 \left(y_2\right)^\dagger\right)$	$+\frac{9g_2^4}{4}$	$+g_Y^2\left(6g_2^2-12\lambda_{S_2}\right)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$8\lambda_{S_2} \text{Tr}\left(y_2 \left(y_2\right)^\dagger\right)$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 6 of 8

Column	Representative term structure
T1	$-9g_2^2\lambda_{S_2}$
T2	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$
T3	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-9g_2^2\lambda_{S_2}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-9g_2^2\lambda_{S_2}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 7 of 8

Column	Representative term structure	
T1	$-12g_Y^2\lambda_{S_2}$	
T2	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$	
T3	$+6\lambda_{12}^{\text{Adj}^2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-12g_Y^2\lambda_{S_2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$	$+6\lambda_{12}^{\text{Adj}^2}$

Terms 8 of 8

Column	Representative term structure
T1	$+6\lambda_{12}^{\text{Cross}2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{\text{Cross}2}$

$\beta_{\lambda_{H1}}^{(1)}$

$16\pi^2\,_{\mu}\frac{d\lambda_{H1}}{d\mu}=\beta_{\lambda_{H1}}^{(1)}$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_{H1}(4\,\text{Tr}\big(y_1\,(y_1)^{\dagger}\big)+6\,\text{Tr}\big(Y_d\,(Y_d)^{\dagger}\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^{\dagger}\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^{\dagger}\big))$
T2	$-4\,\text{Tr}\big(y_1\,(y_1)^{\dagger}\,Y_e\,(Y_e)^{\dagger}\big)$
T3	$+12\lambda_{T3}\,(\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{H1}(4\,\text{Tr}\big(y_1\,(y_1)^{\dagger}\big)+6\,\text{Tr}\big(Y_d\,(Y_d)^{\dagger}\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^{\dagger}\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^{\dagger}\big))$	$-4\,\text{Tr}\big(y_1\,(y_1)^{\dagger}\,Y_e\,(Y_e)^{\dagger}\big)$	$+12\lambda_{T3}\,(\lambda_{T3})^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\lambda_{H1}(2\,\text{Tr}\big(y_1\,(y_1)^{\dagger}\big)+6\,\text{Tr}\big(Y_d\,(Y_d)^{\dagger}\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^{\dagger}\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^{\dagger}\big))$	$-2\,\text{Tr}\big(y_1\,(y_1)^{\dagger}\,Y_e\,(Y_e)^{\dagger}\big)$	$+6\lambda_{T3}\,(\lambda_{T3})^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\lambda_{H1}(3\,\text{Tr}\big(y_1\,(y_1)^{\dagger}\big)+6\,\text{Tr}\big(Y_d\,(Y_d)^{\dagger}\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^{\dagger}\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^{\dagger}\big))$	$-3\,\text{Tr}\big(y_1\,(y_1)^{\dagger}\,Y_e\,(Y_e)^{\dagger}\big)$	$+6\lambda_{T3}\,(\lambda_{T3})^*$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-2\,\text{Tr}\big(y_1\,(y_1)^{\dagger}\,Y_e\,(Y_e)^{\dagger}\big)$	$+4\lambda_{T3}\,(\lambda_{T3})^*$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-2\,\text{Tr}\big(y_1\,(y_1)^{\dagger}\,Y_e\,(Y_e)^{\dagger}\big)$	$+4\lambda_{T3}\,(\lambda_{T3})^*$

Terms 2 of 6

Column	Representative term structure
T1	$-\frac{9g_2^2\lambda_{H1}}{2}$
T2	$-\frac{3g_Y^2\lambda_{H1}}{2}$
T3	$+6\lambda_{12}\lambda_{H2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{9g_2^2\lambda_{H1}}{2}$	$-\frac{3g_Y^2\lambda_{H1}}{2}$	$+6\lambda_{12}\lambda_{H2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-\frac{3g_Y^2\lambda_{H1}}{2}$	$+4\lambda_{12}\lambda_{H2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{3g_Y^2\lambda_{H1}}{2}$	$+4\lambda_{12}\lambda_{H2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H1}}{2}$	$-\frac{3g_Y^2\lambda_{H1}}{2}$	$+2\lambda_{12}\lambda_{H2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H1}}{2}$	$-\frac{3g_Y^2\lambda_{H1}}{2}$	—

Terms 3 of 6

Column	Representative term structure
T1	$+6\lambda_H\lambda_{H1}$
T2	$+4\lambda_{H1}^2$
T3	$+4\lambda_{H1}\lambda_{S1}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$	$+4\lambda_{H1}\lambda_{S1}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$	$+6\lambda_{H1}\lambda_{S1}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$	$+6\lambda_{H1}\lambda_{S1}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$	$+8\lambda_{H1}\lambda_{S1}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$	$+8\lambda_{H1}\lambda_{S1}$

Terms 4 of 6

Column	Representative term structure
T1	$+ \frac{9g_2^4}{4}$
T2	$-9g_2^2 \lambda_{H1}$
T3	$+ \frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+ \frac{9g_2^4}{4}$	$-9g_2^2 \lambda_{H1}$	$+ \frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+ \frac{9g_2^4}{4}$	$-9g_2^2 \lambda_{H1}$	$+ \frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 6

Column	Representative term structure
T1	$\lambda_{H1}(2\text{Tr}\left(y_1\ (y_1)^\dagger\right)+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+4\lambda_{S1}^{\text{Adj}})$
T2	$+6g_2^4$
T3	$+6\lambda_{H1}^{\text{Adj}^2}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\lambda_{H1}(2\text{Tr}\left(y_1\ (y_1)^\dagger\right)+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+4\lambda_{S1}^{\text{Adj}})$	$+6g_2^4$	$+6\lambda_{H1}^{\text{Adj}^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{H1}(2\text{Tr}\left(y_1\ (y_1)^\dagger\right)+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+4\lambda_{S1}^{\text{Adj}})$	$+6g_2^4$	$+6\lambda_{H1}^{\text{Adj}^2}$

Terms 6 of 6

Column	Representative term structure
T1	$+\lambda_{H2}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{H2}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

$\beta_{\lambda_{H2}}^{(1)}$

$16\pi^2\,\mu\,\frac{d\lambda_{H2}}{d\mu}=\beta_{\lambda_{H2}}^{(1)}$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_{H2}(2\,\text{Tr}\big(y_2\,(y_2)^\dagger\big)+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{S_2}^{\text{Adj}})$
T2	$-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$
T3	$+4\lambda_{T3}\,(\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{H2}(2\,\text{Tr}\big(y_2\,(y_2)^\dagger\big)+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{S_2}^{\text{Adj}})$	$-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_{T3}\,(\lambda_{T3})^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_{T3}\,(\lambda_{T3})^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-3\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_{T3}\,(\lambda_{T3})^*$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-4\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+12\lambda_{T3}\,(\lambda_{T3})^*$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{H2}(2\,\text{Tr}\big(y_2\,(y_2)^\dagger\big)+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{S_2}^{\text{Adj}})$	$-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_{T3}\,(\lambda_{T3})^*$

Terms 2 of 6

Column	Representative term structure
T1	$+6g_2^4$
T2	$-\frac{33g_2^2\lambda_{H2}}{2}$
T3	$+3g_Y^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{H2}}{2}$	$+3g_Y^4$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$+3g_Y^4$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$+3g_Y^4$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{9g_2^2\lambda_{H2}}{2}$	$+3g_Y^4$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{H2}}{2}$	$+3g_Y^4$

Terms 3 of 6

Column	Representative term structure
T1	$-\frac{15g_Y^2\lambda_{H2}}{2}$
T2	$+2\lambda_{12}\lambda_{H1}$
T3	$+6\lambda_H\lambda_{H2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{15g_Y^2\lambda_{H2}}{2}$	$+2\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{15g_Y^2\lambda_{H2}}{2}$	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{15g_Y^2\lambda_{H2}}{2}$	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{15g_Y^2\lambda_{H2}}{2}$	$+6\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{15g_Y^2\lambda_{H2}}{2}$	—	$+6\lambda_H\lambda_{H2}$

Terms 4 of 6

Column	Representative term structure
T1	$+4\lambda_{H2}^2$
T2	$+8\lambda_{H2}\lambda_{S2}$
T3	$+6\lambda_{H2}^{\text{Adj}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H2}^2$	$+8\lambda_{H2}\lambda_{S2}$	$+6\lambda_{H2}^{\text{Adj}^2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{H2}^2$	$+6\lambda_{H2}\lambda_{S2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{H2}^2$	$+6\lambda_{H2}\lambda_{S2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+4\lambda_{H2}^2$	$+4\lambda_{H2}\lambda_{S2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H2}^2$	$+8\lambda_{H2}\lambda_{S2}$	$+6\lambda_{H2}^{\text{Adj}^2}$

Terms 5 of 6

Column	Representative term structure
T1	$\lambda_{H2}(2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$
T2	$+ \frac{9g_2^4}{4}$
T3	$-9g_2^2\lambda_{H2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\lambda_{H2}(2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\lambda_{H2}(3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\lambda_{H2}(4\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 6 of 6

Column	Representative term structure
T1	$+ \frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$
T2	$+ \lambda_{H1}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+ \frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+ \frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+ \lambda_{H1}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$

$\beta_{\lambda_{12}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{12}}{d\mu}=\beta_{\lambda_{12}}^{(1)}$

Terms 1 of 9

Column	Representative term structure
T1	$\lambda_{12}(4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+4\lambda_{S_2^{\text{Adj}}})$
T2	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3	$+4\lambda_{T3}\left(\lambda_{T3}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{12}(4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+4\lambda_{S_2^{\text{Adj}}})$	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	$+4\lambda_{T3}\left(\lambda_{T3}\right)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	$+3\lambda_{T3}\left(\lambda_{T3}\right)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$+3\lambda_{T3}\left(\lambda_{T3}\right)^*$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	$+4\lambda_{T3}\left(\lambda_{T3}\right)^*$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	—

Terms 2 of 9

Column	Representative term structure
T1	$-12g_2^2\lambda_{12}$
T2	$-6g_Y^2\lambda_{12}$
T3	$+4\lambda_{12}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-12g_2^2\lambda_{12}$	$-6g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-9g_2^2\lambda_{12}$	$-6g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-9g_2^2\lambda_{12}$	$-6g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-12g_2^2\lambda_{12}$	$-6g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+4\lambda_{12}^2$

Terms 3 of 9

Column	Representative term structure
T1	$+4\lambda_{12}\lambda_{S_1}$
T2	$+8\lambda_{12}\lambda_{S_2}$
T3	$+4\lambda_{H1}\lambda_{H2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{12}\lambda_{S_1}$	$+8\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+8\lambda_{12}\lambda_{S_1}$	$+4\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+4\lambda_{H1}\lambda_{H2}$

Terms 4 of 9

Column	Representative term structure
T1	$\lambda_{12}(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right))$
T2	$+\frac{9g_2^4}{4}$
T3	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\lambda_{12}(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right))$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\lambda_{12}(3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right))$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 9

Column	Representative term structure
T1	$-\frac{9}{2}\text{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$
T2	$\lambda_{12}(2\text{Tr}\left(y_1\ (y_1)^\dagger\right) + 4\text{Tr}\left(y_2\ (y_2)^\dagger\right) + 4\lambda_{S_1}^{\text{Adj}})$
T3	$\lambda_{12}\left(2\text{Tr}\left(y_1\ (y_1)^\dagger\right) + 2\text{Tr}\left(y_2\ (y_2)^\dagger\right) + \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{9}{2}\text{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$\lambda_{12}(2\text{Tr}\left(y_1\ (y_1)^\dagger\right) + 4\text{Tr}\left(y_2\ (y_2)^\dagger\right) + 4\lambda_{S_1}^{\text{Adj}})$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\lambda_{12}\left(2\text{Tr}\left(y_1\ (y_1)^\dagger\right) + 2\text{Tr}\left(y_2\ (y_2)^\dagger\right) + \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}}\right)$

Terms 6 of 9

Column	Representative term structure
T1	$+\lambda_{12}^{\text{Cross}}\left(\frac{4}{3}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\frac{4}{3}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\right)$
T2	$+\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$
T3	$+16g_2^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{12}^{\text{Cross}}\left(\frac{4}{3}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\frac{4}{3}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\right)+\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$		$+16g_2^4$

Terms 7 of 9

Column	Representative term structure
T1	$+g_2^2(-24\lambda_{12}-16\lambda_{12}^{\text{Cross}})$
T2	$+g_Y^2(-6\lambda_{12}-4\lambda_{12}^{\text{Cross}})$
T3	$+\lambda_{S_1}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_2^2(-24\lambda_{12}-16\lambda_{12}^{\text{Cross}})$	$+g_Y^2(-6\lambda_{12}-4\lambda_{12}^{\text{Cross}})$	$+\lambda_{S_1}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$

Terms 8 of 9

Column	Representative term structure
T1	$+\lambda_{S_2}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$
T2	$+16\lambda_{12}^{\text{Adj}^2}$
T3	$+\frac{32\lambda_{12}^{\text{Cross}^2}}{3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{S_2}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$	$+16\lambda_{12}^{\text{Adj}^2}$	$+\frac{32\lambda_{12}^{\text{Cross}^2}}{3}$

Terms 9 of 9

Column	Representative term structure
T1	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$
T2	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$

$\beta_{\lambda_{T3}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{T3}}{d\mu}=\beta_{\lambda_{T3}}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{T3}(2\text{Tr}\left(y_1\ (y_1)^\dagger\right)+\text{Tr}\left(y_2\ (y_2)^\dagger\right)+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+4\lambda_{H2}^{\text{Adj}})$
T2	$-\frac{21g_2^2\lambda_{T3}}{2}$
T3	$-\frac{9g_Y^2\lambda_{T3}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\lambda_{T3}(2\text{Tr}\left(y_1\ (y_1)^\dagger\right)+\text{Tr}\left(y_2\ (y_2)^\dagger\right)+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+4\lambda_{H2}^{\text{Adj}})$	$-\frac{21g_2^2\lambda_{T3}}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{21g_2^2\lambda_{T3}}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{33g_2^2\lambda_{T3}}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$

Terms 2 of 4

Column	Representative term structure
T1	$+2\lambda_{12}\lambda_{T3}$
T2	$+2\lambda_H\lambda_{T3}$
T3	$+4\lambda_{H1}\lambda_{T3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$

Terms 3 of 4

Column	Representative term structure
T1	$+4\lambda_{H2}\lambda_{T3}$
T2	$\lambda_{T3}\left(\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$
T3	$-9g_2^2\lambda_{T3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	$\lambda_{T3}\left(\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$	$-9g_2^2\lambda_{T3}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	$-9g_2^2\lambda_{T3}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—

Terms 4 of 4

Column	Representative term structure	
T1	$\lambda_{T3} \left(\frac{3}{2} \text{Tr} \left(y_1 \ (y_1)^\dagger \right) + \frac{3}{2} \text{Tr} \left(y_2 \ (y_2)^\dagger \right) + 6 \text{Tr} \left(Y_d \ (Y_d)^\dagger \right) + 2 \text{Tr} \left(Y_e \ (Y_e)^\dagger \right) + 6 \text{Tr} \left(Y_u \ (Y_u)^\dagger \right) + \frac{\lambda_{12}^{\text{Adj}}}{6} + \frac{2\lambda_{H1}^{\text{Adj}}}{3} + 2\lambda_{H2}^{\text{Adj}} \right)$	
T2	$\lambda_{T3} (\text{Tr} \left(y_1 \ (y_1)^\dagger \right) + 2 \text{Tr} \left(y_2 \ (y_2)^\dagger \right) + 6 \text{Tr} \left(Y_d \ (Y_d)^\dagger \right) + 2 \text{Tr} \left(Y_e \ (Y_e)^\dagger \right) + 6 \text{Tr} \left(Y_u \ (Y_u)^\dagger \right) - 4\lambda_{H1}^{\text{Adj}})$	
T3	$\lambda_{T3} \left(\text{Tr} \left(y_1 \ (y_1)^\dagger \right) + \text{Tr} \left(y_2 \ (y_2)^\dagger \right) + 6 \text{Tr} \left(Y_d \ (Y_d)^\dagger \right) + 2 \text{Tr} \left(Y_e \ (Y_e)^\dagger \right) + 6 \text{Tr} \left(Y_u \ (Y_u)^\dagger \right) - \frac{4\lambda_{12}^{\text{Adj}}}{3} - 2\lambda_{12}^{\text{Cross}} - 2\lambda_{H1}^{\text{Adj}} - \frac{4\lambda_{H2}^{\text{Adj}}}{3} \right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\lambda_{\tau 3} \left(\frac{3}{2} \text{Tr} \left(y_1 \ (y_1)^\dagger \right) + \frac{3}{2} \text{Tr} \left(y_2 \ (y_2)^\dagger \right) + 6 \text{Tr} \left(Y_d \ (Y_d)^\dagger \right) + 2 \text{Tr} \left(Y_e \ (Y_e)^\dagger \right) + 6 \text{Tr} \left(Y_u \ (Y_u)^\dagger \right) + \frac{\lambda_{12}^{\text{Adj}}}{6} + \frac{2\lambda_{H1}^{\text{Adj}}}{3} + 2\lambda_{H2}^{\text{Adj}} \right)$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$\lambda_{\tau 3} (\text{Tr} \left(y_1 \ (y_1)^\dagger \right) + 2 \text{Tr} \left(y_2 \ (y_2)^\dagger \right) + 6 \text{Tr} \left(Y_d \ (Y_d)^\dagger \right) + 2 \text{Tr} \left(Y_e \ (Y_e)^\dagger \right) + 6 \text{Tr} \left(Y_u \ (Y_u)^\dagger \right) - 4\lambda_{H1}^{\text{Adj}})$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\lambda_{\tau 3} \left(\text{Tr} \left(y_1 \ (y_1)^\dagger \right) + \text{Tr} \left(y_2 \ (y_2)^\dagger \right) + 6 \text{Tr} \left(Y_d \ (Y_d)^\dagger \right) + 2 \text{Tr} \left(Y_e \ (Y_e)^\dagger \right) + 6 \text{Tr} \left(Y_u \ (Y_u)^\dagger \right) - \frac{4\lambda_{12}^{\text{Adj}}}{3} - 2\lambda_{12}^{\text{Cross}} - 2\lambda_{H1}^{\text{Adj}} - \frac{4\lambda_{H2}^{\text{Adj}}}{3} \right)$

$\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{12}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{12}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+2\text{Tr}\big(y_2\ (y_2)^\dagger\big))$
T2	$-\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$
T3	$+\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\lambda_{12}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+2\text{Tr}\big(y_2\ (y_2)^\dagger\big))$	$-\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\lambda_{12}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+3\text{Tr}\big(y_2\ (y_2)^\dagger\big))$	$-\frac{2}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+\frac{2}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$	$-\frac{2}{9}\lambda_{T3}\ (\lambda_{T3})^*$

Terms 2 of 5

Column	Representative term structure
T1	$-9g_2^2\lambda_{12}^{\text{Adj}}$
T2	$-6g_Y^2\lambda_{12}^{\text{Adj}}$
T3	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-9g_2^2\lambda_{12}^{\text{Adj}}$	$-6g_Y^2\lambda_{12}^{\text{Adj}}$	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-9g_2^2\lambda_{12}^{\text{Adj}}$	$-6g_Y^2\lambda_{12}^{\text{Adj}}$	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-24g_2^2\lambda_{12}^{\text{Adj}}$	$-6g_Y^2\lambda_{12}^{\text{Adj}}$	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$

Terms 3 of 5

Column	Representative term structure
T1	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$
T2	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$
T3	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	—

Terms 4 of 5

Column		Representative term structure
T1	$\lambda_{12}^{\text{Adj}}$	$\left(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$
T2	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$	
T3	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{12}^{\text{Adj}}\left(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$		$+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$

Terms 5 of 5

Column	Representative term structure
T1	$+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

$\beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{12}^{\text{Cross}}}{d\mu}=\beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{12}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+8\lambda_{12}^{\text{Cross}}+\lambda_{S_1}^{\text{Adj}}+\lambda_{S_2}^{\text{Adj}}\right)$
T2	$+\lambda_{12}^{\text{Cross}}\left(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\right)$
T3	$-\frac{7}{6}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{12}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+8\lambda_{12}^{\text{Cross}}+\lambda_{S_1}^{\text{Adj}}+\lambda_{S_2}^{\text{Adj}}\right)+\lambda_{12}^{\text{Cross}}\left(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\right)$	$-\frac{7}{6}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	

Terms 2 of 5

Column	Representative term structure
T1	$-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^{*}$
T2	$+\frac{33g_2^4}{4}$
T3	$+g_2^2\left(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+\frac{33g_2^4}{4}$	$+g_2^2\left(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}}\right)$

Terms 3 of 5

Column	Representative term structure
T1	$+g_Y{}^2\left(-\frac{3\lambda_{12}}{2}-6\lambda_{12}^{\text{Cross}}\right)$
T2	$+\lambda_{12}{}^2$
T3	$+\lambda_{S_1}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_Y{}^2\left(-\frac{3\lambda_{12}}{2}-6\lambda_{12}^{\text{Cross}}\right)$	$+\lambda_{12}{}^2$	$+\lambda_{S_1}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$

Terms 4 of 5

Column	Representative term structure
T1	$+\lambda_{S_2}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$
T2	$+4\lambda_{12}^{\text{Adj}^2}$
T3	$+11\lambda_{12}^{\text{Cross}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{S_2}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$	$+4\lambda_{12}^{\text{Adj}^2}$	$+11\lambda_{12}^{\text{Cross}^2}$

Terms 5 of 5

Column	Representative term structure
T1	$-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$
T2	$-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$
T3	$+\lambda_{H1}\lambda_{H2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$	$-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$	$+\lambda_{H1}\lambda_{H2}$

$\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{H1}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{H1}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$
T3	$+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\lambda_{H1}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\lambda_{H1}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$

Terms 2 of 4

Column	Representative term structure
T1	$-9g_2^2\lambda_{H1}^{\text{Adj}}$
T2	$-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$
T3	$+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-9g_2^2\lambda_{H1}^{\text{Adj}}$	$-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-9g_2^2\lambda_{H1}^{\text{Adj}}$	$-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$	—

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$
T2	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$
T3	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$

Terms 4 of 4

Column	Representative term structure
T1	$\lambda_{H1}^{\text{Adj}} (2 \text{Tr} \left(y_1 \left(y_1 \right)^{\dagger} \right) + 6 \text{Tr} \left(Y_d \left(Y_d \right)^{\dagger} \right) + 2 \text{Tr} \left(Y_e \left(Y_e \right)^{\dagger} \right) + 6 \text{Tr} \left(Y_u \left(Y_u \right)^{\dagger} \right) + 16 \lambda_{S1}^{\text{Adj}})$
T2	$-\frac{33 g_2^2 \lambda_{H1}^{\text{Adj}}}{2}$
T3	$+12 \lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\lambda_{H1}^{\text{Adj}} (2 \text{Tr} \left(y_1 \left(y_1 \right)^{\dagger} \right) + 6 \text{Tr} \left(Y_d \left(Y_d \right)^{\dagger} \right) + 2 \text{Tr} \left(Y_e \left(Y_e \right)^{\dagger} \right) + 6 \text{Tr} \left(Y_u \left(Y_u \right)^{\dagger} \right) + 16 \lambda_{S1}^{\text{Adj}})$	$-\frac{33 g_2^2 \lambda_{H1}^{\text{Adj}}}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{H1}^{\text{Adj}} (2 \text{Tr} \left(y_1 \left(y_1 \right)^{\dagger} \right) + 6 \text{Tr} \left(Y_d \left(Y_d \right)^{\dagger} \right) + 2 \text{Tr} \left(Y_e \left(Y_e \right)^{\dagger} \right) + 6 \text{Tr} \left(Y_u \left(Y_u \right)^{\dagger} \right) + 16 \lambda_{S1}^{\text{Adj}})$	$-\frac{33 g_2^2 \lambda_{H1}^{\text{Adj}}}{2}$	$+12 \lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$

$\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{H2}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+16\lambda_{S2}^{\text{Adj}})$
T2	$+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$
T3	$+\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+16\lambda_{S2}^{\text{Adj}})$	$+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+16\lambda_{S2}^{\text{Adj}})$	$+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{8}{9}\lambda_{T3}\ (\lambda_{T3})^*$

Terms 2 of 5

Column	Representative term structure
T1	$+g_Y^2\left(4g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$
T2	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$
T3	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+g_Y^2\left(4g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+g_Y^2\left(12g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+g_Y^2\left(12g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_Y^2\left(4g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$

Terms 3 of 5

Column	Representative term structure
T1	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$
T2	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$
T3	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	$\lambda_{H2}^{\text{Adj}}(3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—

Terms 4 of 5

Column	Representative term structure
T1	$+8\,\mathrm{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$
T2	$+16\lambda_{T3}\,(\lambda_{T3})^*$
T3	$-9g_2^2\lambda_{H2}^{\mathrm{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+8\,\mathrm{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+16\lambda_{T3}\,(\lambda_{T3})^*$	$-9g_2^2\lambda_{H2}^{\mathrm{Adj}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-4\,\mathrm{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+16\lambda_{T3}\,(\lambda_{T3})^*$	$-9g_2^2\lambda_{H2}^{\mathrm{Adj}}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$
T2	$+12\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+12\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$

$\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_1}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+13\lambda_{S_1}^{\text{Adj}}\right)$
T2	$+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)$
T3	$-\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+13\lambda_{S_1}^{\text{Adj}}\right)$	$+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)$	$-\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+13\lambda_{S_1}^{\text{Adj}}\right)$	$+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)$	$-\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$

Terms 2 of 5

Column	Representative term structure
T1	$+ \frac{3g_2^4}{4}$
T2	$+g_2^2(-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$
T3	$+ \frac{\lambda_{12}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$	$+ \frac{\lambda_{12}^2}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$	—

Terms 3 of 5

Column	Representative term structure
T1	$+\frac{\lambda_{H1}^2}{2}$
T2	$+3\lambda_{H1}^{\text{Adj}2}$
T3	$+\frac{7\lambda_{S1}^2}{4}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{\lambda_{H1}^2}{2}$	$+3\lambda_{H1}^{\text{Adj}2}$	$+\frac{7\lambda_{S1}^2}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{\lambda_{H1}^2}{2}$	$+3\lambda_{H1}^{\text{Adj}2}$	$+\frac{7\lambda_{S1}^2}{4}$

Terms 4 of 5

Column	Representative term structure
T1	$+22\lambda_{S_1}^{\text{Adj}^2}$
T2	$+\frac{3\lambda_{12}^2}{4}$
T3	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+22\lambda_{S_1}^{\text{Adj}^2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+22\lambda_{S_1}^{\text{Adj}^2}$	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$

Terms 5 of 5

Column	Representative term structure
T1	$+12\lambda_{12}^{\text{Adj}^2}$
T2	$-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_{12}^{\text{Adj}^2}$	$-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$

$\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_2}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$
T2	$+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$
T3	$-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$	$+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$	$-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$	$+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$	$-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$

Terms 2 of 6

Column	Representative term structure
T1	$+ \frac{3g_2^4}{4}$
T2	$+g_Y^2 \left(9g_2^2 - \frac{3\lambda_{S_2}}{2} - 12\lambda_{S_2}^{\text{Adj}} \right)$
T3	$+g_2^2 (-3\lambda_{S_2} - 24\lambda_{S_2}^{\text{Adj}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	$+g_Y^2 \left(9g_2^2 - \frac{3\lambda_{S_2}}{2} - 12\lambda_{S_2}^{\text{Adj}} \right)$	$+g_2^2 (-3\lambda_{S_2} - 24\lambda_{S_2}^{\text{Adj}})$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	$+g_Y^2 \left(9g_2^2 - \frac{3\lambda_{S_2}}{2} - 12\lambda_{S_2}^{\text{Adj}} \right)$	$+g_2^2 (-3\lambda_{S_2} - 24\lambda_{S_2}^{\text{Adj}})$

Terms 3 of 6

Column	Representative term structure
T1	$+ \frac{3g_Y^4}{2}$
T2	$+ \frac{\lambda_{12}^2}{4}$
T3	$+ \frac{\lambda_{H2}^2}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+ \frac{3g_Y^4}{2}$	$+ \frac{\lambda_{12}^2}{4}$	$+ \frac{\lambda_{H2}^2}{2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \frac{3g_Y^4}{2}$	—	$+ \frac{\lambda_{H2}^2}{2}$

Terms 4 of 6

Column	Representative term structure
T1	$+3\lambda_{H2}^{\text{Adj}^2}$
T2	$+\frac{7\lambda_{S2}^2}{4}$
T3	$+22\lambda_{S2}^{\text{Adj}^2}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$	$+22\lambda_{S2}^{\text{Adj}^2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$	$+22\lambda_{S2}^{\text{Adj}^2}$

Terms 5 of 6

Column	Representative term structure
T1	$+\frac{3\lambda_{12}^2}{4}$
T2	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3	$+12\lambda_{12}^{\text{Adj}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$	$+12\lambda_{12}^{\text{Adj}^2}$

Column	Representative term structure
T1	$-\frac{\lambda_{12}^{\text{Cross}2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\lambda_{12}^{\text{Cross}2}}{2}$